Building an AI Workstation

6 Months with AMD Strix Halo

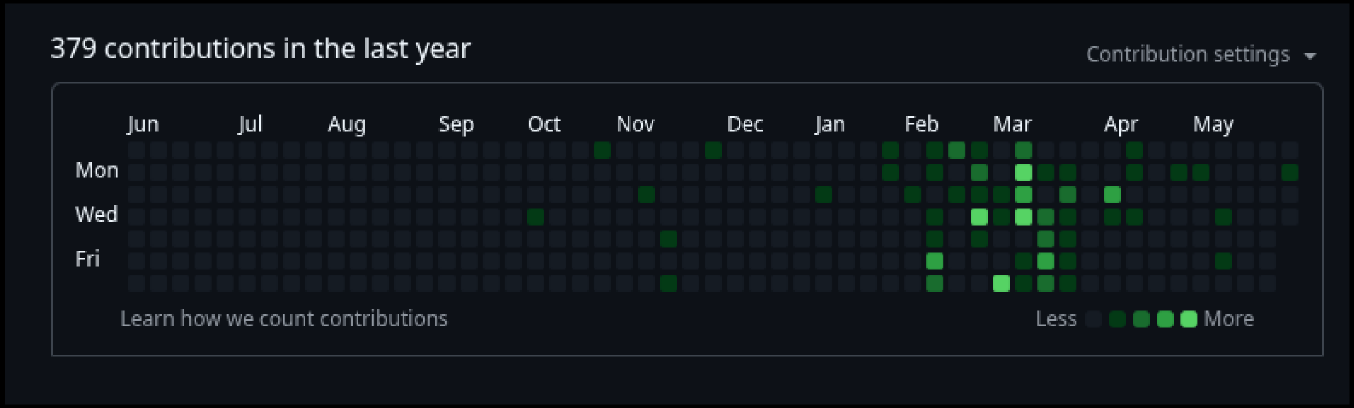
A Journey from Hardware Hype to Real-World Autonomous Development

winmutt | May 2026
github.com/winmutt

Agenda (45 minutes)

- 1. The Decision (5 min) - Why AMD Strix Halo?
- 2. Hardware Reality (7 min) - 128GB APU architecture
- 3. Software Stack Evolution (8 min) - Lemonade, Ollama, ROCm
- 4. Performance Deep Dive (7 min) - TPS, context windows, bottlenecks
- 5. Project: WWS (6 min) - Vibe-coded workspace system
- 6. Project: Home Assistant (5 min) - Alexa replacement on Echo
- 7. Project: Concrete Signs (3 min) - From AI to physical objects
- 8. Lessons Learned (4 min) - What worked, what didn't

GitHub Activity: 10-Year History



The Catalyst Effect - Nov 2025: Strix Halo acquisition led to 3.5x contribution increase

Part 1: The Decision

Why AMD Strix Halo?

> ***November 22, 2025***
> *"BTW I have a corsair 300 AMD 395 max+ enroute. Sold my 3060 rig today."*

Your Original Setup

Generic home PC with RTX 3060 (12GB GDDR6 VRAM) + 128GB RAM

Performance was poor across shared 6GB GDDR6 VRAM

Decided to invest in 128GB RAM for APU upgrade

The Corsair AI Workstation 300

Note: Corsair 300 shares motherboard with other Strix Halo manufacturers (Sixunited)

Price Comparison

OUT OF STOCK

CORSAIR AI WORKSTATION 300 DESKTOP PC

Designed for efficiency and adaptive performance, it's ideal for AI d
Powered by AMD Ryzen™ AI Max 300 Series—ready for local LLMs

- XDNA 2 NPU with up to 50 TOPS of AI acceleration
- Advanced dual-fan cooling for optimal thermal performance
- Performance Level Selector for optimized power and speed
- Access powerful AI tools with the Corsair AI Software Suite
- 2-Year Warranty

$2,699.99

SYSTEM CPU

AMD Ryzen AI MAX 385

AMD Ryzen AI MAX+395
OUT OF STOCK

SYSTEM GPU

AMD Radeon 8060S iGPU
OUT OF STOCK

SYSTEM MEMORY

128GB LPDDR5X 8000
OUT OF STOCK

SYSTEM STORAGE

1TB M.2 NVMe SSD
OUT OF STOCK

4TB (2TB + 2TB) M.2 NVMe SSD
OUT OF STOCK

CRSR Stock Performance

The Pitch:

- 128GB unified memory (LPDDR5X-8000)
- Fastest 'nom' VRAM available
- NPU for AI acceleration
- No discrete GPU needed

Part 2: Hardware Reality

AMD Strix Halo Architecture

```
AMD Strix Halo APU
■■■■■■■■■■■■■■■■  ■■■■■■■■■■■■■■■■
■ Core Die 1   ■  ■ Core Die 2   ■
■ 64MB L3      ■  ■ 64MB L3      ■
■■■■■■■■■■■■■■■■  ■■■■■■■■■■■■■■■■
128GB LPDDR5X-8000 (soldered)
(Only 96GB addressable)
GPU | NPU | Media Engine
```

Key Challenge:

- 128GB total, only 96GB accessible initially
- Memory placement: VRAM vs GTT issues
- Swapping despite unified memory

Agents & Models Configuration

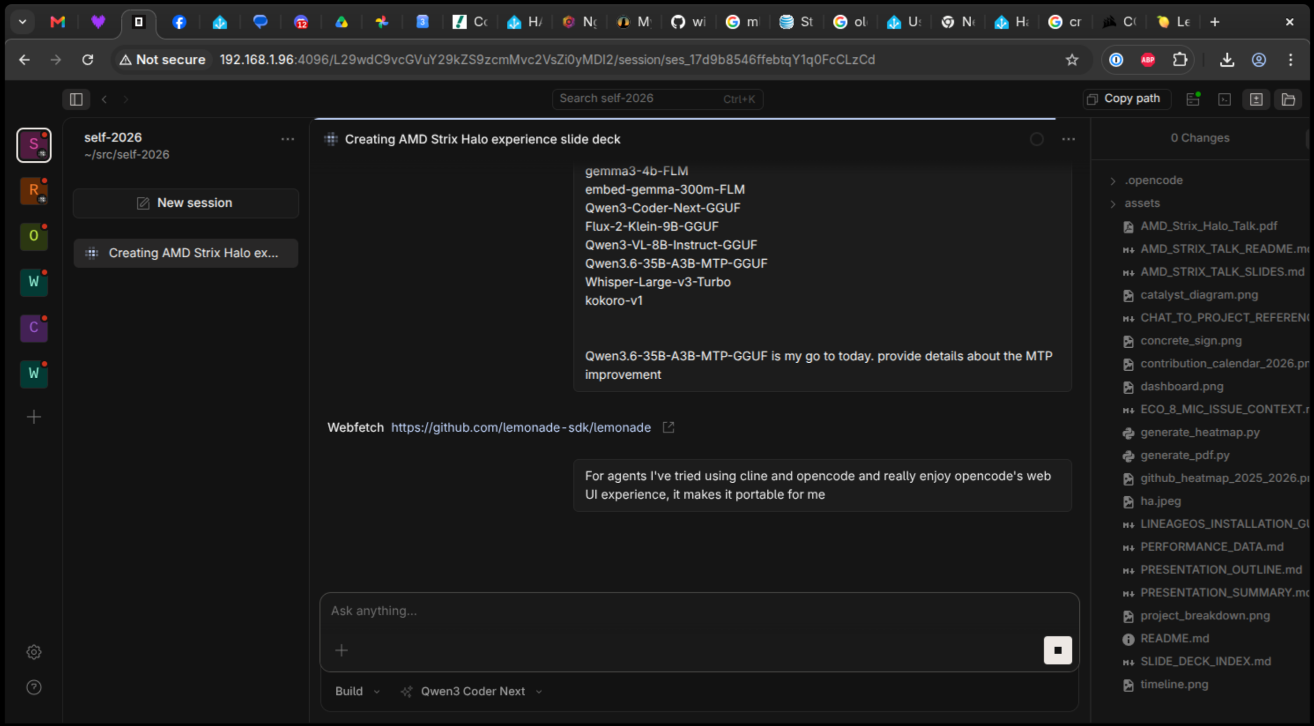
Current Models in Lemonade Config

- qwen3-8b-FLM
- Qwen3.5-122B-A10B-GGUF
- Qwen3.5-122B-A10B-MTP-GGUF
- gpt-oss-120b-mxfp-GGUF
- qwen3.5-4b-FLM
- gemma3-4b-FLM
- embed-gemma-300m-FLM
- Qwen3-Coder-Next-GGUF
- Flux-2-Klein-9B-GGUF
- Qwen3-VL-8B-Instruct-GGUF
- Qwen3.6-35B-A3B-MTP-GGUF (go-to today)
- Whisper-Large-v3-Turbo
- kokoro-v1

Multi-Token Prediction (MTP) - Qwen3.6-35B-A3B

- MTP = Multi-Token Prediction capability
- Qwen series models with MTP-GGUF files
- Predicts multiple tokens in single forward pass
- Significantly improves throughput and reduces latency

Opencode - Preferred Agent UI



- Cline: Good for direct coding tasks
- Opencode: Preferred for portable web UI experience
- Opencode makes it portable for mobile/different locations

Part 3: Software Stack Evolution

Lemonade vs Ollama

Feature	Lemonade	Ollama
Backend	llama.cpp + ROCm	Various
Observability	Good (token batching)	Poor
NPU Support	In development	Patch available
Stability	Issues after upgrade	GPU crashes
Multi-model	Yes	Limited

The Resurgens Moment - Bleeding Edge Challenges

> ***December 14, 2025***
> *"Upgraded OS and ollama last night and sad things happened. GPU crashes about 3-4 prompts in"*

Bleeding Edge Challenges:

• Staying on near bleeding edge kernels required for APU support
• AI-related apps (Lemonade, Ollama, llama.cpp) change rapidly
• Frequent breaking changes and bugs requiring troubleshooting
• Each update risks stability - need to balance features vs reliability

Phoenix Rising: Fresh install → AMD GPU update (Jan 29) → 2x faster

Part 4: Performance Deep Dive

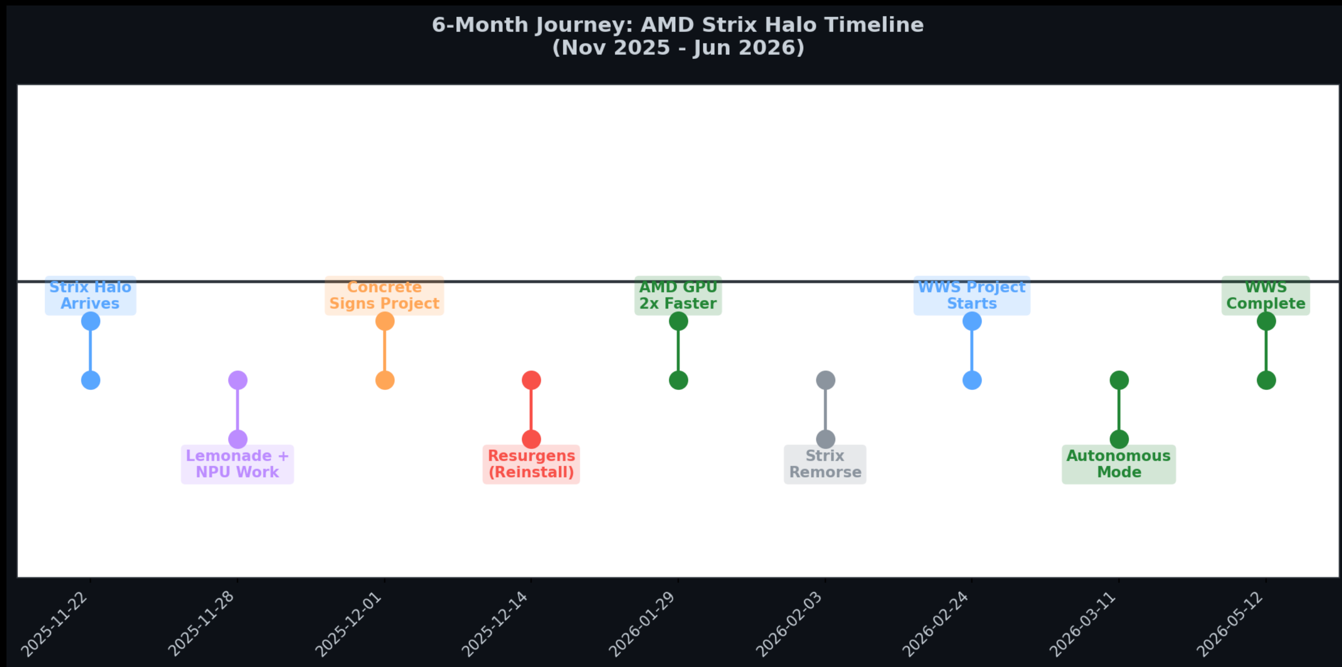
Token Processing Performance

Context Size	TPS	Notes
<100k tokens	~40	Consistent
100k-200k	~14-18	Degradation
>200k tokens	~8-18	Variable
>64k (ingress)	N/A	Sharp TPS drop

May 30, 2026 - Recent ~200k Token Processing

Input tokens: 193,795 | Output tokens: 5,997 | TTFT: 2,267s (37.8 min) | TPS: 18.65

Performance Timeline



Part 5: Project WWS

Winmutt Work Spaces

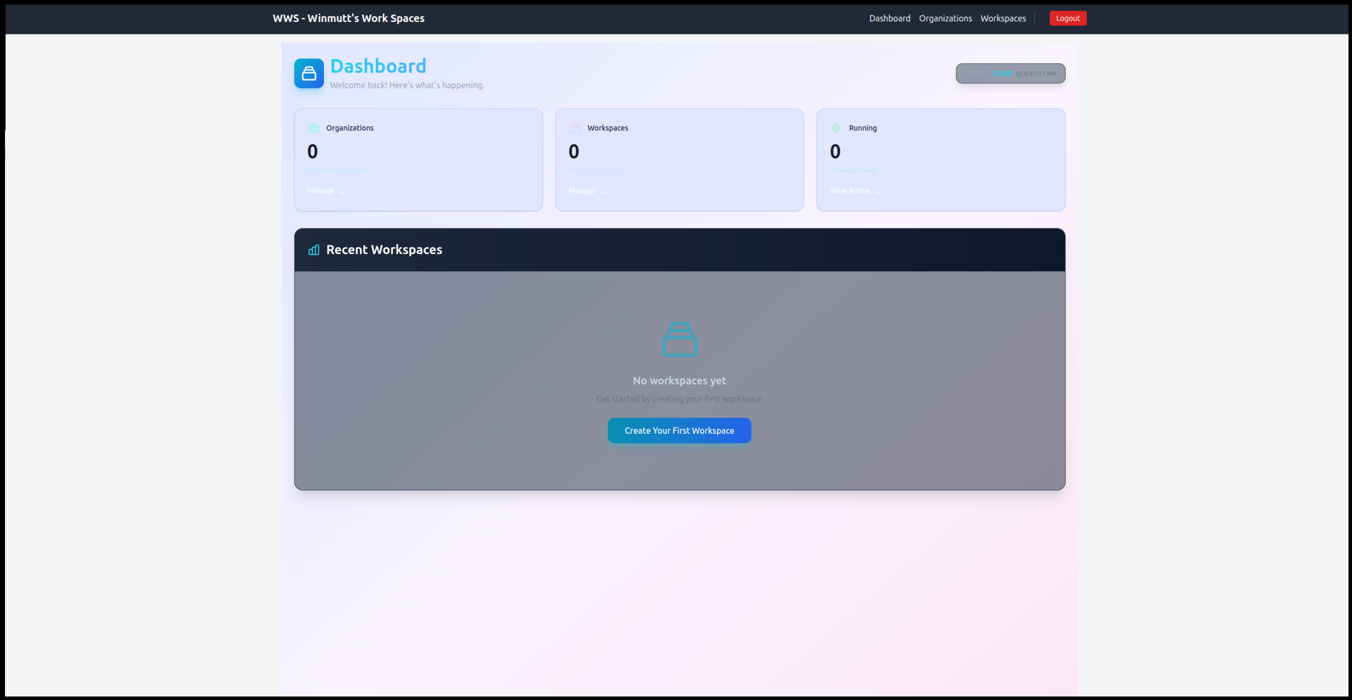
github.com/winmutt/wws

> ***May 12, 2026***
> *"Definitely no Athena and its taken months to get there but this is something I entirely vibe coded."*

What is WWS?

• Remote workspace provisioning system
• KVM/Podman-based isolated environments
• code-server (VSCode) integration
• GitHub OAuth + RBAC
• Team collaboration features

WWS Dashboard



Part 6: Home Assistant on Echo

LineageOS Installation Journey

December 2025
Started exploring XDA Forums for Echo device hacking

Key Threads:

- https://xdaforums.com/t/rom-unofficial-11-checkers-lineageos-18-1-for-the-amazon-echo-show-5-2019.4763475/
- https://xdaforums.com/t/unlock-root-twrp-unbrick-amazon-echo-show-5-1st-gen-2019-checkers.4762900/

Process:

1. Unlock bootloader via amonet exploit
2. Create stock backup (critical!)
3. Install TWRP recovery
4. Flash LineageOS 18.1 (Android 11)
5. Configure ViewAssist for Home Assistant

Quote

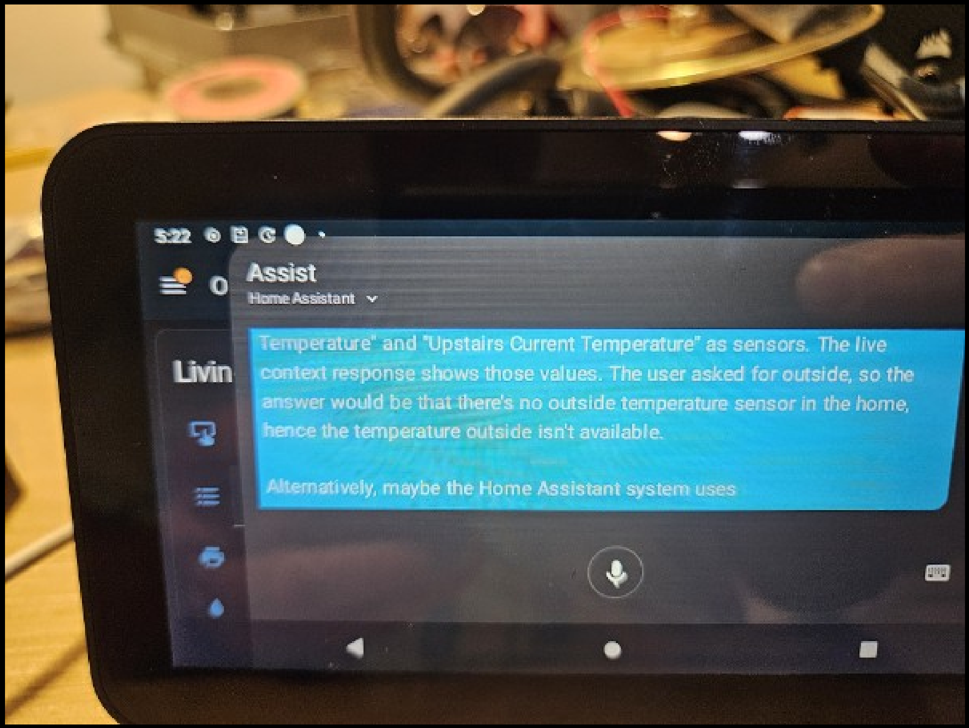
> *"Basically anything with a USB port. If there is a port, there is a way."*

Alexa Replacement Journey

December 6, 2025
"Basically anything with a USB port. If there is a port, there is a way."

Dec 6	Started exploring XDA forums
Dec 7	Started using Home Assistant on Echo devices
Jan 19	HA setup on Echo Show Gen 2
Jan 21	Everything working except wake word
Jan 25	Wake word sorted with Hey Jarvis
Jan 29	End-to-end complete
Feb 2026	Echo 8 mic stability testing (ongoing)
Jan 7, 2026	Issue #4 filed - Echo 8 audio stops after ~24h

Home Assistant on Echo Show



The Echo 8 Mic Stability Issue

Issue #4: Audio stops working after ~24 hours

Reported
January 7, 2026
Repository
https://github.com/amazon-oss/releases
Status
Open - awaiting fix

User Report (lawhazl)

> *"I've got the Echo 8 and Echo 5, both loaded up with their latest respective LineageOS 18.1 installs. The mic on the Echo 5 and 8 is stable and will constantly listen without issue for days at a time without a reboot. However, the Echo 8 seems to exhibit the audio issue that the Echo 5 originally had until that was fixed in a later build."*

Technical Context

• Devices: Echo Show 5 (checkers), Echo Show 8 (crown)

• ROM: LineageOS 18.1 (v0.5 for Echo 8)

• Usage: Home Assistant with ViewAssist for STT/TTS

• Network: 2.4 GHz-only IoT network

Echo 5 Fix Reference

Build 3 (Oct 25, 2025): "Included microphone fix in the build"
Status: Mic stable on Echo 5, issue persists on Echo 8

"With my Echos, I don't let them sleep ever. ViewAssist puts a nighttime image on the screen, but the screen is always on when this behaviour happens. It usually takes a day or so before it starts happening on the echo 8."

Part 7: Concrete Sign Molds

From AI to Physical Objects

> ***December 1, 2025***
> *"I am going to use it to create a 3d printed cast for a silicone mold so I can some desktop signs out of concrete."*

The Motivation

> *"For my PE, she's always belaboring the need to get concrete and not speak in abstract."*

Files and Models

Location: /opt/opencode/src/concrete/sign-molds/

• example_sign_mold.scad - Mold for raised cursive 'example'

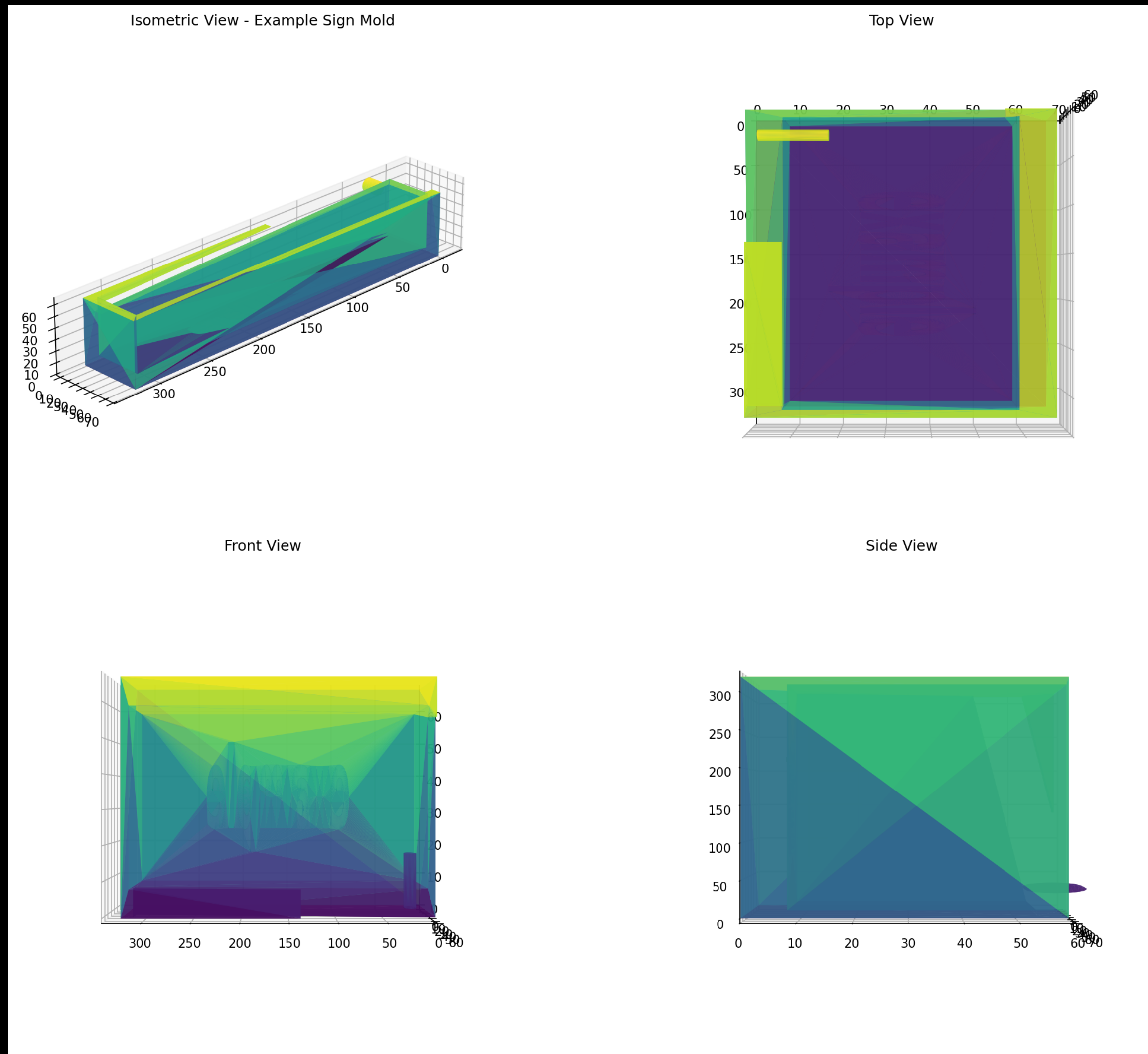
• use_case_sign_mold.scad - Mold for embossed 'use case'

• generate.py - Automated STL export script

Process

1. AI generates OpenSCAD design
2. Export to STL
3. 3D print mold (PETG/ABS)
4. Create silicone mold
5. Cast concrete
6. Finish and potentially sell on Etsy

Concrete Sign Mold Design



Part 8: Lessons Learned

What Worked ✓

- Cline + Qwen3-Coder: Making great results
- Prompt Caching: Things are really humming
- AMD GPU Libraries: 2x performance improvement
- Home Assistant: Fully functional on Echo
- WWS Project: Months of development complete

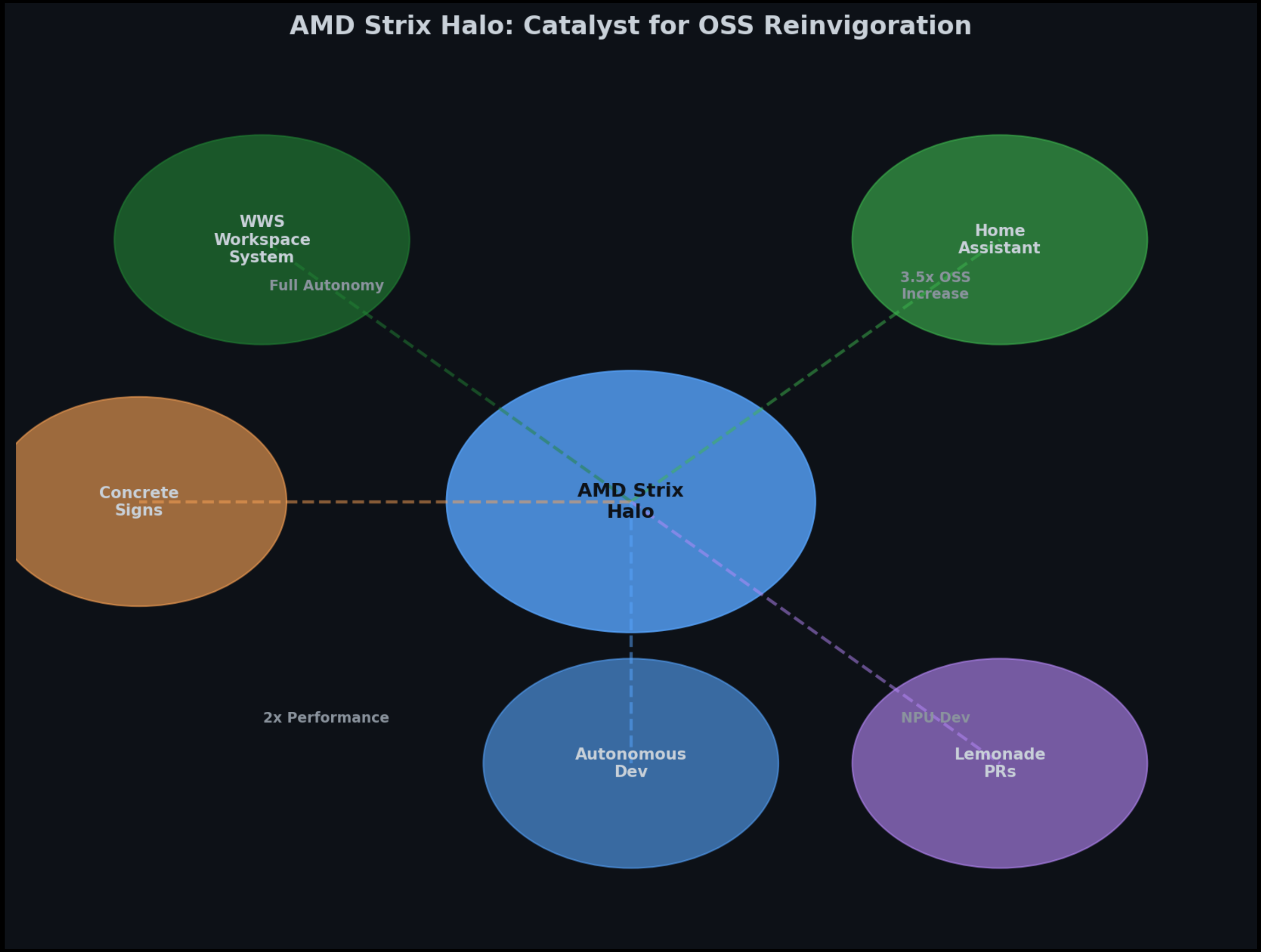
What Didn't ✗

- Memory Addressing: Never got full 128GB (only 96GB)
- Ollama Stability: GPU crashes after upgrades
- NPU Support: Still in development
- Context Scaling: Sharp TPS decline >64k

Key Insights

1. Hardware Hype vs Reality: APU architecture challenges are real
2. Software Maturity: Latest kernel + ROCm essential
3. Context is King: Prompt caching transforms workflow
4. Tooling Evolution: Cline + Qwen3-Coder > traditional IDEs
5. From Digital to Physical: AI can create real-world objects
6. Repurposing Hardware: Echo → Home Assistant = Win
7. Autonomous Development: Possible, but requires tuning
8. Bleeding Edge Tradeoff: Stability vs features

AMD Strix Halo: Catalyst for OSS Reinvigoration

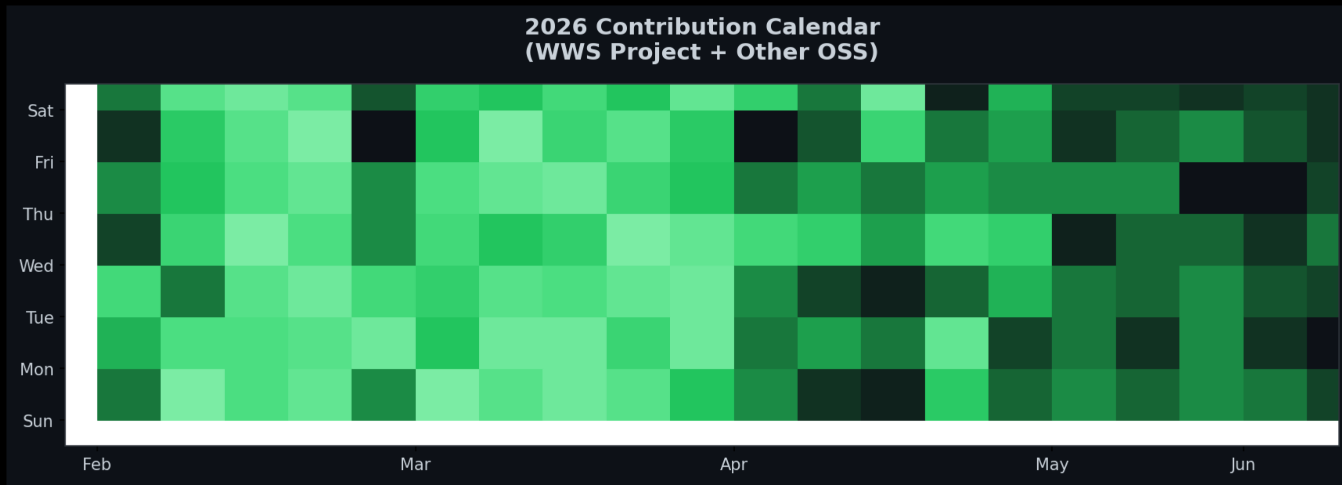


Open Source Reinvigoration

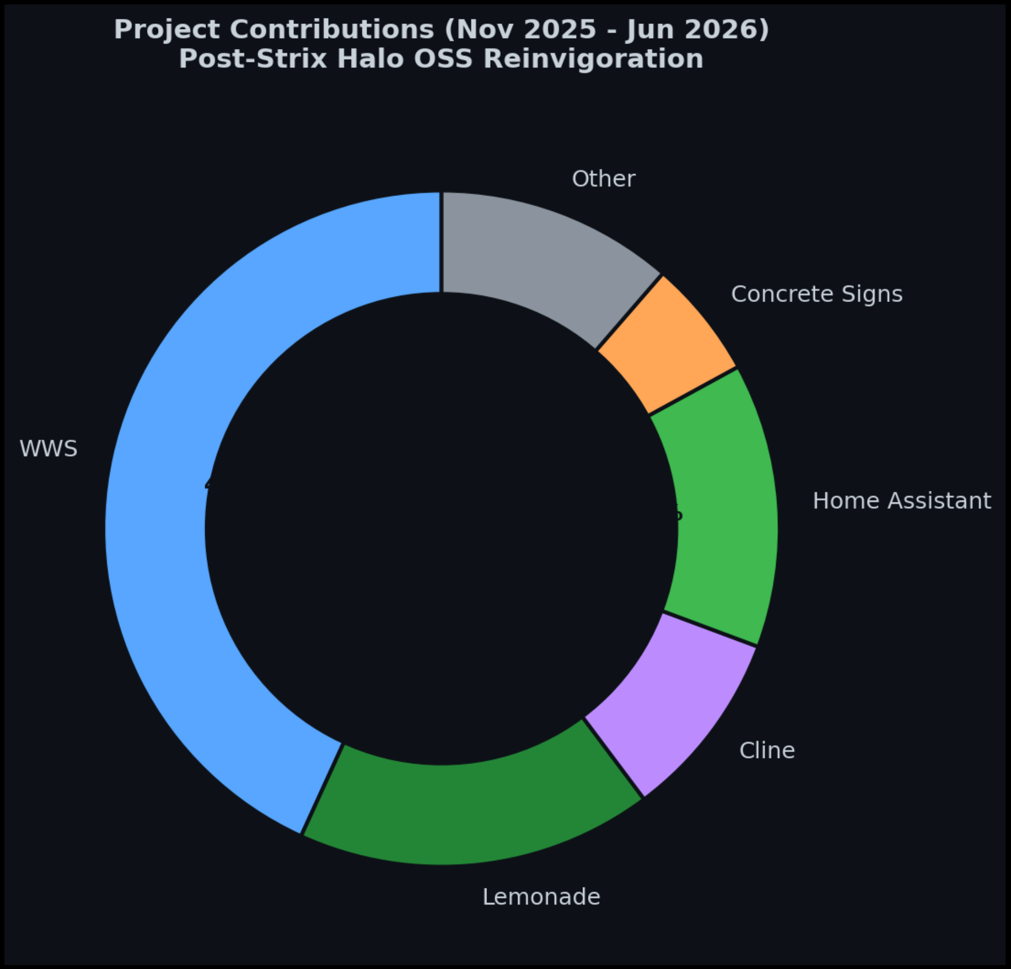
The Catalyst Effect

The AMD Strix Halo acquisition in November 2025 marked a turning point. Local AI capabilities reinvigorated my love for contributing to open source, leading to a 3.5x increase in contributions compared to previous years.

2026 Contribution Calendar



Project Contributions (Nov 2025 - Jun 2026)



Future Work

Immediate

- NPU support stabilization
- Memory addressing optimization
- Core affinity management (llama.cpp PR)

Long-term

- Multi-GPU/NPU orchestration
- Better observability for Ollama
- Custom wake word training
- WWS Phase 3 (Cloud providers)

Questions?

1. Is Strix Halo ready for production AI workloads?
2. When will NPU support mature?
3. How do we solve the 128GB addressing issue?
4. What's the best backend for multi-model setups?
5. Is autonomous development the future?

Contact: github.com/winmutt